

Refutation of Conjecture 00000004413

Cycle counts in bounded-degree graph limits

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1 The conjecture

Definition 1 (graphing / bounded-degree graph limit). *A (deterministic) graphing on a finite probability space V is a symmetric, irreflexive edge relation E on V . It satisfies degree bound d when every vertex has at most d neighbours. Its cycle measure of length l is the number of distinct undirected l -cycles divided by $|V|$ (the uniform measure on V).*

Conjecture 00000004413. *For every graphing of degree bound d and every l , the l -cycle measure is at most $\frac{(d-1)^l}{2l}$, and this bound is attained and optimal for the d -regular tree limit.*

2 The conjecture is false

2.1 Counterexample to the upper bound

Take $G = K_3$, the triangle, on the probability space $V = \{0, 1, 2\}$ ($x \sim y \iff x \neq y$).

- Degree bound: every vertex has degree 2, so $d = 2$.
- The number of undirected 3-cycles is 1, so the 3-cycle measure is $1/3$.
- The claimed bound: $\frac{(d-1)^l}{2l} = \frac{1^3}{6} = \frac{1}{6}$.

But $\frac{1}{3} > \frac{1}{6}$: the alleged upper bound fails on the simplest bounded-degree object. Already the raw cycle *count* 1 exceeds $1/6$.

2.2 The optimality/attainment claim is also false

A d -regular *tree* contains no cycles at all, so its l -cycle measure is 0, whereas the claimed bound $\frac{(d-1)^l}{2l}$ is strictly positive for every $d \geq 2$ and $l \geq 1$. Hence the bound is never “attained” by the d -regular tree: the second half of the conjecture is inconsistent with the first.

2.3 What $(d-1)^l/(2l)$ really is

The expression $(d-1)^l/(2l)$ is the *mean of the Poisson limit* for the number of l -cycles in a uniformly random d -regular graph on n vertices (Bollobás–Wormald): the cycle count of a random d -regular graph converges in distribution to $\text{Poisson}((d-1)^l/(2l))$. It is therefore a *typical/expected* value for locally-tree-like limits, not a universal upper bound.

3 Formalization

The Lean 4/Mathlib development (`lean4/Main.lean`) defines

- `Graphing V`: symmetric irreflexive decidable relation on a finite vertex type V (uniform measure);
- `Graphing.deg`, `Graphing.DegreeBound`, `Graphing.cycleWalks`, `Graphing.edgeSetOf` (a `Finset (Sym2 V)`), so a cycle and all its rotations/reversal give the same edge set), `Graphing.cycleCount` (normalized by $|V|$), and `Graphing.claimedBound` $= (d - 1)^l / (2l)$;
- `triangle`: the K_3 graphing on `Fin 3`.

Main theorems:

- `conjecture_00000004413_false`: the universal bound is *false* (proved from `triangle_degreeBound`, `triangle_cycleCount_ge` $\Rightarrow$ `cycleCount` $\geq 1/3$, and `claimedBound_2_3` $= 1/6$);
- `counterexample_exists`: explicit counterexample;
- `tree_cannot_attain`: `claimedBound` > 0 for $d \geq 2$, $l \geq 1$, while a tree has cycle measure 0, so the bound cannot be attained by the d -regular tree.

The axiom audit (`lean4/Check.lean`, `#print axioms`) reports dependence only on Lean’s three standard foundational axioms `propext`, `Classical.choice`, `Quot.sound` — no sorry, no `native_decide`.

4 Reproduction

- `reproduce.py`: brute-force enumeration of undirected 3-cycles in K_3 ($= 1$), comparison with $1^3/6 = 1/6$, and a check that trees have cycle count $0 < (d - 1)^l / (2l)$.
- `lean4/`: lake build exits 0 on Lean v4.33.1/Mathlib v4.33.1.